

Title

Exercise-induced modulation of the FGF21-adiponectin axis in older women: differential responses by metabolic phenotype

Abstract

Background: The fibroblast growth factor-adiponectin axis integrates systemic metabolic signals, but exercise-induced adaptations may vary based on initial metabolic status.

Objective: To determine how 10 weeks of supervised exercise modulates FGF21, adiponectin, and their ratio in older women stratified by metabolic syndrome presence. **Methods:** A total of 60 women (mean age 75.3 years) with (n=32) or without (n=28) metabolic syndrome were randomized to supervised combined exercise (resistance plus elastic-band training, 3x/week) or usual-activity control. Circulating biomarkers, body composition, fitness, and metabolic parameters were assessed at baseline and week 10. **Results:** Exercise increased adiponectin in both metabolic groups (non-metabolic syndrome effect size $d=1.02$; metabolic syndrome $d=0.68$). FGF21 responses differed: increased in the non-metabolic syndrome group ($d=0.40$) but decreased in the metabolic syndrome group ($d=-0.31$). The FGF21-to-adiponectin ratio declined in both exercise groups, with larger reduction in metabolic syndrome participants ($d=-0.84$ versus $d=-0.28$). Secondary improvements included reduced body fat, improved maximal oxygen uptake, decreased fasting glucose, and favorable lipid changes in both intervention groups (all $P < 0.05$). **Conclusion:** Exercise induces differential FGF21 responses depending on metabolic phenotype, potentially reflecting adaptive metabolic reprogramming in metabolic syndrome.